Preface

Abstract: This project uses machine learning, deep learning, and data scraping techniques to study public sentiment in the context of the 2024 Indian election. A thorough grasp of public opinion is imperative given the quickly changing political scene, especially in this era of social media and numerous outlets for the diffusion of information. Our dual objectives are to: (1) develop a sentiment analysis model specifically tailored for the 2024 election to provide real-time insights into public sentiment; and (2) evaluate the model's performance and accuracy to understand the impact of sentiment on political discourse. To achieve this, we will collect a diverse array of text data from sources such as news articles, blogs, forums, and social media platforms, ensuring a balanced representation of positive, negative, and neutral political viewpoints.

The model will be trained on labeled datasets using text tokenization techniques and will incorporate both machine learning (ML) and deep learning (DL) models, including LSTM and BERT. The effectiveness of the model will be measured using evaluation metrics such as accuracy, precision, recall, and F1 score. Additionally, real-time analysis tools will be implemented to capture and reflect dynamic changes in sentiment throughout the election period. This initiative aims to provide stakeholders with valuable insights into public opinion, enabling them to make informed decisions and better understand the electoral landscape.